

Effects of added connective tissues on the sensory and mechanical properties of restructured beef steaks.

To quantify objectionable levels of connective tissues, restructured beef products were made with 2.5 and 5% added tendon; 5 and 10% added epimysium, gristle, or peri/endomysium; and a control. Initial tenderness (IT), residual connective tissue (CT), and overall texture (OT) were evaluated by a sensory panel. Panelists adversely scored IT, CT, and OT for 2.5 and 5% tendon and CT and OT for 10% epimysium and gristle. CT and OT scores correlated with hydroxyproline content and Lee-Kramer peak shear force for uncooked steaks with added tendon, gristle and epimysium but not peri/endomysium. Acceptable products can be made when raw materials are free of tendons and contain only limited amounts of epimysium. Copyright © 1990. Published by Elsevier Ltd.